ZOO507 - Principles of Animal Ecology

Grand Quiz Preparation

(Part-1)

Topic 1-46

Environmental Science

THE ENVIRONMENT is everything around us. It includes all of the **living and the nonliving things** with which
We are utterly dependent on the environment for air, water, food, shelter, energy.

ENVIRONMENTAL SCIENCE, an **interdisciplinary** study of how humans interact with the environment of living and nonliving things. It integrates information and ideas from the **natural sciences**, such as biology, chemistry, and geology, the **social sciences**, *such as geography, economics, political* science, and demography, and the ***humanities***, including philosophy and ethics

Major Fields of Study Related to Environmental Science

Major Fields	Subfields
Biology: study of living things (organisms)	**Ecology:** study of how organisms interact with one another and with their nonliving environment
	Botany: study of plants
	Zoology: study of animals
Chemistry: study of chemicals and their interactions	**Biochemistry:** study of the chemistry of living things
Earth science: study of the planet as a whole and its nonliving systems	**Climatology:** study of the earth's atmosphere and climate
	Geology: study of the earth's origin, history, surface, and interior processes
	Hydrology: study of the earth's water resources
	Paleontology: study of fossils and ancient life
Social sciences: studies of human society	**Anthropology:** study of human cultures
	Demography: study of the characteristics of human populations
	Geography: study of the relationships between human populations and the earth's surface features
	Economics: study of the production, distribution, and consumption of goods and services
	Political Science: study of the principles, processes, and structure of government and political institutions
Humanities: study of the aspects of the human condition not covered by the physical and social sciences	**History:** study of information and ideas about humanity's past
	Ethics: study of moral values and concepts concerning right and wrong human behavior and responsibilities
	Philosophy: study of knowledge and wisdom about the nature of reality, values, and human conduct

Environmental science is an interdisciplinary study of connections between the earth's life-support system and **human activities**.

GOALS OF ENVIRONMENTAL SCIENCE

The goals of environmental science are to learn

How nature works?
How the environment affects us?
How we affect the environment?
How to deal with environmental problems and live more sustainably?

A key **subfield of environmental science** is **ecology**, the biological science that studies *how organisms, or living things, interact with their **environment** and with **each other***.

Every organism is a member of a certain **species**: a group of **organisms with distinctive traits** and, for sexually reproducing organisms, can mate and produce fertile offspring.
For example, all **humans** are members of a species that biologists have named ***Homo sapiens*.**

A major **focus of ecology** is the study of **ecosystems**.
AN ECOSYSTEM is a **set of organisms** interacting with **one another** and with their **environment** of **nonliving matter** and **energy** within a defined area or volume.

We should not confuse environmental science and ecology with **environmentalism, a social movement** dedicated ***to protecting the earth's life-support systems*** for us and all other forms of life.

Environmentalism is practiced more in the political and ethical arenas than in the realm of science.

ENVIRONMENTAL SUSTAINABILITY

Sustainability is the ability of the **earth's** various **natural systems and human cultural systems and economies** *to survive and adapt to changing environmental conditions indefinitely*

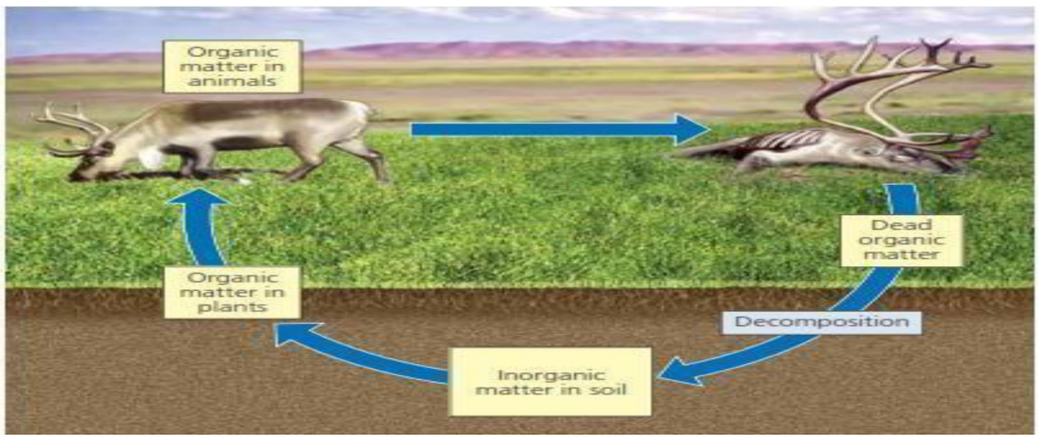
Sustainability
A critical component of sustainability is **natural capital—the natural *resources* and natural *services* that** keep us and other forms of life alive and support our economies.

1. **Natural resources** are ***materials*** and ***energy*** in nature that are essential or useful to humans

These resources are often classified as **renewable** (such as air, water, **soil**, plants, and wind) or ***nonrenewable*** (such as **copper**, oil, and coal).

2. **Natural services** are functions of nature, such as ***purification of air*** and ***water***, which support life and human economies.

Ecosystems provide us with these essential services at **no cost**.
One vital natural service is **nutrient cycling,** the ***circulation of chemicals*** necessary for life, from the environment (mostly from soil and water) through organisms and back to the environment



Nutrient cycling: an important **natural service** that recycles chemicals needed by organisms from the environment (mostly from soil and water) through organisms and back to the environment. **Natural capital** is supported by **solar capital**: energy from the sun Take away solar energy, and all natural capital would collapse.

SOLAR ENERGY

Solar energy **warms the planet** and supports **photosynthesis**—a complex chemical process that plants use to provide food for themselves and for us and most other animals.

• This direct input of solar energy also produces **indirect forms** of renewable solar energy such as wind, flowing water, and biofuels made from plants and plant residues.
• For example, ***topsoil****, the upper layer of the earth's crust,* provides the nutrients that support the plants, animals, and microorganisms that live on land; when they die and decay, they resupply the soil with these nutrients. Without this service, life as we know it could not exist.
• Thus, **our lives and economies** depend on energy from the **sun** (solar capital) and **natural resources** and **natural services** (natural capital) provided by the earth

SECOND COMPONENT OF SUSTAINABILITY

A second component of sustainability is **to recognize that many human activities** can *degrade natural capital* by using normally renewable resources **faster** than nature can renew them
For example, in parts of the world, we are **clearing mature forests** much faster than nature can replenish them
We are also **harvesting many species of ocean fish** faster than they can replenish themselves This leads us to a third component of sustainability
Environmental scientists search for *solutions to problems* such as the degradation of natural capital
However, their work is **limited to finding the scientific solutions**, while the political solutions are left to **political processes**
For example, scientific solutions might be to stop chopping down biologically diverse, mature forests, and to harvest fish no faster than they can replenish themselves
But implementing such solutions could require government laws and regulations

The search for solutions often involves **conflicts**

GROWTH OF ENVIRONMENTALLY SUSTAINABLE SOCIETIES

Economic growth is an increase in a nation's **output** of goods and services

It is usually measured by the **percentage of change** in a country's **gross domestic product (GDP)**: **the annual market value** of all goods and services produced by all firms and organizations, foreign and domestic, operating within a country

Changes in a country's economic growth **per person** are measured by **per capita GDP**:
the **GDP divided by the total population** at **midyear**.

Consumers in the rich country are said to **have more purchasing power** than consumers in the second country have

To help compare countries, economists use a tool called **purchasing power parity (PPP)**
By combining **per capita GDP** and **PPP**, for any given country, they arrive at a **per capita GDP PPP**—a measure of the amount of goods and services that a country's **average citizen could buy**

While **economic growth** provides people with more goods and services, **economic development** has the goal of using economic growth **to improve living standards**.

The United Nations classifies the world's countries as **economically developed** or **developing** based primarily on their degree of industrialization and their per capita GDP PPP.

The **developed countries** (with 1.2 billion people) include the United States, Canada, Japan, Australia, New Zealand, and most countries of Europe
Most are highly industrialized and have a high per capita GDP PPP

All other nations (with 5.5 billion people) are classified as **developing countries**, most of them in Africa, Asia, and Latin America

Some are middle-income, **moderately developed** countries such as China, India, Brazil, Turkey, Thailand, and Mexico.
Others are low-income, **least developed** countries where per capita GDP PPP is steadily declining

Some economists call for continuing **conventional economic growth**, which has helped to **increase food supplies**, allowed people to live longer, and **stimulated mass production** of an array of useful goods and services for many people

- They also see such growth **as a cure for poverty**, maintaining that some of the resulting increase in wealth trickles down to countries and people near the bottom of the economic ladder

Other economists put much greater emphasis on **environmentally sustainable economic** development

AFFECTS OF ECOLOGICAL FOOTPRINTS ON THE EARTH

- From a human standpoint, **a resource** is **anything obtained from the environment to meet our needs and wants**
- **Conservation** is the **management of natural resources** with the goal of minimizing resource waste and sustaining resource supplies for current and future generations
- Some resources, such as **solar energy, fresh air, wind, fresh surface water**, fertile soil, and wild edible plants, are **directly available for use**
- Other resources such as **petroleum, iron, water found underground**, and cultivated crops, are **not directly available**

They **become useful** to us only with **some effort and technological ingenuity**.

For example, **petroleum** was a **mysterious fluid** until we learned how to find, extract, and convert (refine) it into gasoline, heating oil, and other products that could be sold

AFFECTS OF ECOLOGICAL FOOTPRINTS ON THE EARTH

Solar energy is called a **perpetual resource** because it is **renewed continuously** and is expected to last **at least 6 billion years** as the sun completes its life cycle

On a human time scale, a **renewable resource** can be **replenished fairly quickly** (from hours to hundreds of years) through natural processes as long as it is not used up faster than it is renewed

Examples include **forests, grasslands, fisheries, freshwater, fresh air, and fertile soil**

- The **highest rate** at which a **renewable resource can be used indefinitely** without reducing its **available supply** is called its **sustainable yield**
- When we **exceed** a **renewable resource's natural replacement rate**, the available supply begins to shrink, a process known as **environmental degradation**

OVEREXPLOITATION OF COMMONLY SHARED RENEWABLE RESOURCES

There are **three types of property** or **resource rights**
1- One is **private property** where **individuals or firms own the rights** to land, minerals, or other resources

2- Another is **common property** where the rights to certain resources **are held by large groups of individuals**

3- A third category consists of ***open access renewable re-sources***, ***owned by no one and available for use by anyone* at little or no charge**

Many **common property and open access** renewable resources have been **degraded**

In 1968, biologist Garrett Hardin (1915–2003) called such degradation the ***tragedy of the commons***

It occurs *because each user* of a shared common resource or open-access resource reasons, "**If I do not use this resource, someone else will**"

OVEREXPLOITATION OF COMMONLY SHARED RENEWABLE RESOURCES

Such **resource degradation** results from the **push to satisfy the short-term needs** and wants of a growing number of people

It threatens our ability to ensure the long-term economic and environmental sustainability of open-access resources such as clean air or an open ocean fishery

- One solution is to use **shared resources** at rates well below their estimated sustainable yields by reducing use of the resources, regulating access to the resources, or doing both

For example, the most common approach is for governments **to establish laws and regulations** limiting the annual harvests of various types of ocean fish that are being harvested at unsustainable levels in their coastal waters

- Another approach is for nations to **enter into agreements** that regulate access to open access renewable resources such as the fish in the open ocean
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NONRENEWABLE RESOURCES

Nonrenewable resources **exist in a fixed quantity**, or stock, in the earth's crust
On a time scale of **millions to billions of years**, geological processes can renew such resources

But on the much shorter human time scale of **hundreds to thousands of years**, these resources can be **depleted** much faster than they are formed

Such **EXHAUSTIBLE RESOURCES** include **energy resources** (such as coal and oil), metallic **mineral resources** (such as copper and aluminum), and **nonmetallic mineral resources** (such as salt and sand)

HOW TO REDUCE DEPENDENCE?

- A **mix of renewable energy** resources such as **wind, the sun, flowing water**, and the heat in the earth's interior could reduce our dependence on nonrenewable fossil fuels such as oil and coal
- Some nonrenewable resources, such as copper and aluminum, can be recycled or reused to extend supplies
- **Recycling** involves **collecting waste materials and processing** them into new materials
- But energy resources such as **oil and coal cannot be recycled** once burned, their energy is no longer available to us

Recycling nonrenewable metallic resources takes much **less energy**, water, and other resources and produces much **less pollution** and **environmental degradation** than exploiting virgin metallic resources

Reusing such resources takes **even less energy and other resources** and produces less pollution and environmental degradation than recycling does

In 2003, humanity total or global ecological footprint was **about 25% higher** than the earth ecological capacity (bottom) and is projected to be **twice** the planet ecological capacity by **2050**.

OUR ECOLOGICAL FOOTPRINTS ARE GROWING

The **per capita ecological footprint** is the average ecological footprint of an individual in a given country or area.

If a country's, or the world's, total ecological footprint is larger than its biological capacity to replenish its renewable resources and absorb the resulting waste products and pollution, it is said to have an **ecological deficit**.

The World Wildlife Fund (WWF) and the Global Footprint Network estimated that in 2003 (the latest data available) humanity's global ecological footprint exceeded the earth's biological capacity by about 25%
The per capita ecological footprint is an estimate of how much of the earth's renewable resources an individual consumes

After the oil-rich United Arab Emirates, the United States has the world's second largest per capita ecological footprint

In 2003, its per capita ecological footprint was about 4.5 times the average global footprint per person, 6 times larger than China's per capita footprint, and 12 times the average per capita footprint in the world's low-income countries like Pakistan

According to William Rees and Mathis Wackernagel, the developers of the ecological footprint concept, it would take the land area of about five more planet earths for the rest of the world to reach current U.S. levels of consumption with existing technology

Put another way, if everyone consumed as much as the average American does today, the earth's natural capital could support only about 1.3 billion people—not

In other words, we are living unsustainably by depleting and degrading some of the earth's irreplaceable natural capital and the natural renewable income it provides as our ecological footprints grow and spread across the earth's surface

INTRODUCTION TO ECOLOGY

The word 'ecology' was first used by **Ernest Haeckel** in **1869**.
Ecology is the scientific study of the interactions that determine the distribution and abundance of organisms in an environment.
All organisms have to interact with both living and non-living things that surround them.
Both these living and non-living factors contribute to the whole environment of organisms in which they live.
Ecology is a complex and an amazing field of biology that describes the richness of the biosphere.

The basic concept of ecology is that, "all organisms in an environment are interdependent".
So environment is a main part of the relationship that occurs between organism and their natural and manmade environment.

Major components of ecology are

- • Species
- • Population
- • Community
- • Ecosystem
- • Biome
- • Biosphere

Sub fields of ecology

- • Organismal ecology
- • Population ecology
- • Community ecology
- • Ecosystem ecology
- • Landscape ecology
- • Global ecology

ECOSYSTEM

The word ecosystem was coined by **Tansley in 1935.**

An ecosystem is a **community of different species** interacting with one another and with their nonliving environment of soil, water, other forms of matter, and energy, mostly from the sun.
Ecosystems can be **natural or artificial** (human created). Examples of **artificial ecosystems** are **crop fields, tree farms, and reservoirs**.
Ecosystems **do not have clear boundaries** and are not isolated from one another. Birds and various other species migrate from one ecosystem to another.

Ecosystem is a basic **structural and functional unit of ecology**.
Structure of an ecosystem depends on **species diversity** as well as the energy that is required to maintain it.

If there is **more species diversity** then structure of ecosystem will be **more complex** and **less energy** will be needed to maintain the structure of ecosystem.

An ecosystem consists of two major components
1. Biotic components
2. Abiotic components

Biotic components
Living components of an ecosystem
1. Producers : Green plants
2. Consumers: Animals
3. Decomposers: Bacteria and fungi

Abiotic components
Non-living components of an ecosystem

Physical components:
1. Water
2. light
3. Temperature
4. Wind
5. Air
6. soil

Chemical components
1. Oxygen
2. Carbon
3. Nitrogen
4. Phosphorus
5. Sulphur.

HOW LIVING SYSTEMS ARE SUSTAINED

All living systems from a cell to the biosphere are dynamic systems that are constantly changing in response to changing environmental conditions.

The **composition of communities and ecosystems changes** due to

- **Continents movement**
- **Climate changes**
- **Natural disturbances**
- **Succession**

Living systems contain complex networks of **positive and negative feedback loops** that interact to provide some degree of stability, or sustainability, over each system's expected life span.

A positive feedback loop causes a system to change further in the **same direction** from which it is moving.
For example when vegetation is removed from a stream valley, flowing water from precipitation causes erosion and loss of nutrients, which causes more vegetation to die. With even less vegetation to hold soil in place, flowing water causes even more erosion and nutrient loss, which causes even more plants to die.

A negative, or corrective, feedback loop causes a system to **change in the opposite direction** from which it is moving. A simple example is a **thermostat**, a device that controls how often, and how long a heating or cooling system runs.

It is useful to distinguish among **two aspects of stability in living systems**.

1. Inertia, or persistence
The ability of a living system, such as a grassland or a forest**, to survive in conditions of moderate disturbances**.

2.Resilience
The **ability of a living system to be restored through secondary succession** after a moderate disturbance

Level of organization: Cell

Cell is the basic structural and functional unit of life.
The cell is the simplest collection of matter that can live. **Cell structure** is correlated to **cellular function.**
Organisms are divided into **two types** on the basis of their **cell structure**.

- Prokaryotes
- Eukaryotes

Prokaryotes consist of **only unicellular organisms** while **Eukaryotes consist of both unicellular and multicellular organisms**.

Prokaryotes
They lack well defined nucleus and also membrane bound organelles.
Eukaryotes
They have well defined nucleus and also have Membrane Bound organelles.

SPECIES

A group of sexually reproducing organisms that can **mate and produce fertile offspring**

The species is the **largest unit of population** in which **gene flow is possible**.
It is defined by reproductive isolation from other species in natural environments.

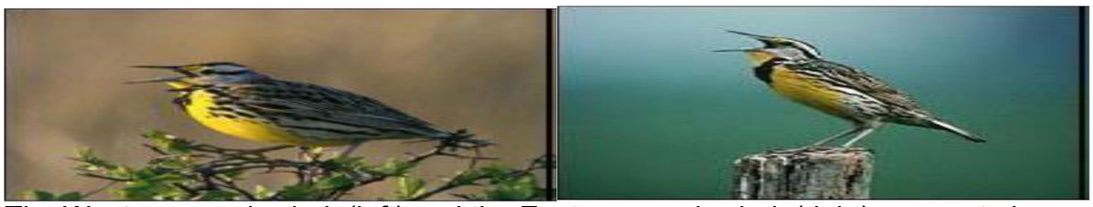
Four basic concepts about species are
1. Biological species concept
2. Ecological species concept

3. Morphological species concept
4. Phylogenetic species concept

1. Biological concept of species

Defines species on the base of their **interbreeding potential** not on similarity or appearance.

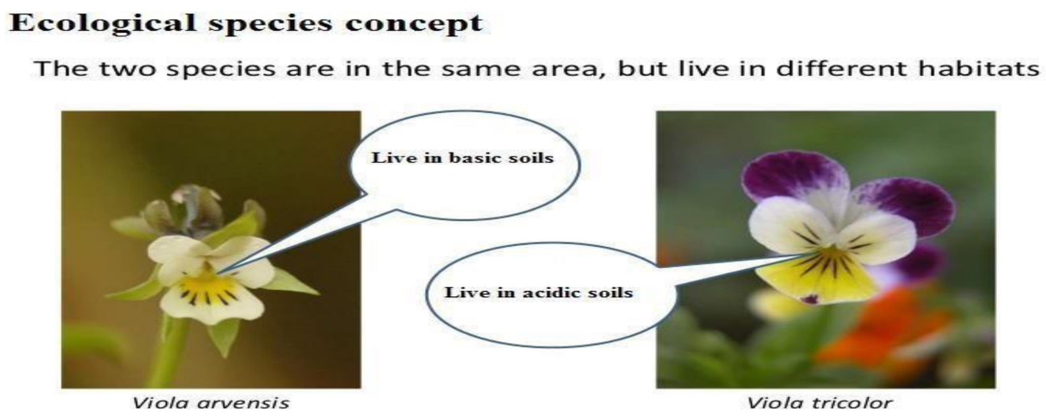
For example, Western meadowlarks (*Sturnellaneglecta*) and Eastern meadowlarks (*Sturnella magna*) look almost identical to one another, yet do not interbreed with each other—thus, they are separate species according to this definition.



The Western meadowlark (left) and the Eastern meadowlark (right) appear to be identical, and their ranges overlap, but their distinct **songs** prevent interbreeding.

2. Ecological species concept

Defines species in terms of **its ecological niche**.



3. Morphological species concept

Defines species by **structural features**.



4. Phylogenetic species concept

Defines species as the smallest group of individuals on a **phylogenetic tree**.



POPULATION

A group of same species located at the same place and same time is called as a **population**. Population biologists study groups in a **specific area**.

A group of ducks would not be a population if there were **mallard and wood ducks combined.**
It means that there are **two populations existing in the same area**.
Every population has particular characteristics

- Population size
- Population growth
- Population distribution
- Population density

Organisms are **born** and the **population size increases**.
Other factors that will increase the size of a population are

- **Reproductive fitness**
- **Good weather**
- **Medicine**
- **Geographic isolation**
- **No predators**

Organisms will also die of old age and the **population will decrease**.
Populations can also decrease when there are **more predators** around, after a natural disaster, when **competition** is too great or when some of the organisms just leave the area (**migration**). Other reasons

Physical limits
There are some **physical limits for a population**. Sometimes other **environments** are just bad places to live for a population. They may be too hot, too cold, or too dangerous.

Geographic limits
Geographic limits are also a main constraints that **decreases the size of a population**.
These above mentioned factors (Physical limits and geographic limits) are very important to address because they limit the population expansion.

COMMUNITY

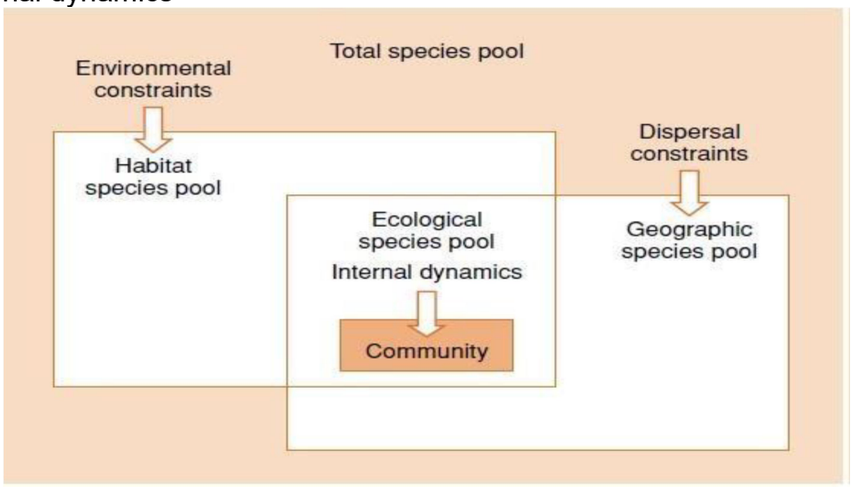
Community or biological community is the **group of population of different species located in a particular area or place.**

Communities are not only a random composition of species but they also have a unique organization.

A community is made up of **many populations** that interact in an area.
For example, a catfish species in a pond usually shares the pond with other fish species, and with plants, insects, ducks, and many other species that make up the community.
Many of the organisms in a community interact with one another in feeding and other relationship.
Communities, like all biological entities, require matter for their construction and energy for their activities.
The community is an assemblage of species populations that occur together in space and time.

Community assembly
The species that assemble to make up a community are determined by:

- Dispersal constraints
- Environmental constraints
- Internal dynamics



The relationships among five types of species pools: **the total pool of species in a region**, **the geographic pool** (species able to **arrive at a site**), the **habitat pool** (species able to persist **under the abiotic conditions** of the site), the **ecological pool** (the overlapping set of species that can **both arrive and persist**) and the **community** (the pool that remains in the face of biotic interactions).

The ecological niche
The **ecological niche** is an important concept of community structure.

An important **principle of ecology** is that each **species has a distinct role to play** in the ecosystems where it is found.
So, **ecological niche** is defined as the **role that a species plays** in its environment including behaviour and influence.

A species' niche should **not be confused with its habitat**, which is the place where it lives.

Niche is simply a **pattern of living of species**.

The ecological niche basically highlights the **functional position of a species** in its environment.

A typical niche describes;
1. The **habitat** which is the actual place of living of an organisms.
2. The **activity pattern** of an organism that is a time period in which an organism remains active.

3. The **resources** that an organism gains from the surrounding environment.

Bacteria from the genus Pseudomonas can grow in a broad variety of niches, which range from natural environments to human-associated ecosystems.

GENERALIST SPECIES

Species that have **broad range of niches** are known as generalist species.
They can live in many different places, eat a variety of foods, and often tolerate a wide range of environmental conditions.
Flies, cockroaches, mice, rats, white-tailed deer, raccoons, and humans are Generalist species.

Cockroaches: Nature's Ultimate Survivors

Cockroaches are present on earth since **350 million years**.
There are **3500 cockroach species**. They can eat almost anything, including algae, dead insects, fingernail, glue, Paper soap. They can also live and breed almost anywhere except in Polar Regions.
• Some cockroach species can go for a **month without food**, survive for a month on a drop of water from a dishrag, and withstand massive doses of radiation.

• One species can survive being **frozen for 48 hours.**

• They have **high reproductive rates**.

In only a year, a **single Asian cockroach** and its offspring can add about **10 million** new cockroaches to the world.

• About **25 species** of cockroach **live in homes** and can carry viruses and bacteria that cause diseases.

SPECIALIST SPECIES

Those species that **occupy narrow range of niches** are known as specialist species.
• They may be able to live in only one type of **habitat**, use one or a few types of **food**, or tolerate a narrow range of climatic and other **environmental conditions**.

• This makes specialists **more prone to extinction** when environmental conditions change.

For example, **tiger salamanders** breed only in **fishless ponds** where their larvae will not be eaten.
Some **shorebirds** occupy specialized niches, feeding on crustaceans, insects, and other organisms on sandy beaches and their adjoining coastal wetlands.
This specialization **Reduces competition and allows sharing of limited resources**.

Is it better to be a generalist or a specialist?
It depends on environmental conditions.

When environmental **conditions are fairly constant**, as in a tropical rain forest, **specialists have an advantage** because they have fewer competitors.

But under **rapidly changing environmental conditions**, the **generalist** usually is better off than the specialist.
China's giant panda is highly endangered because of a combination of habitat loss, low birth rate, and its specialized diet consisting mostly of bamboo.

NATIVE SPECIES

Native species are those species that **normally live and thrive in a particular ecosystem** because **of its soil, isolation and climatic conditions**.
Native species **occurs naturally within a region**. They either evolve in that region or arrive and become established in that region **without human assistance**.
Native species can be either **endemic or indigenous**.

1. Endemic species

These species are **found within a specific part** of world only.

Blind Indus dolphin is only found in Indus river of Pakistan.
Red kangaroo is endemic to Australia only in the World

2. Indigenous species

These are **found within the region and elsewhere in the world** also.

Snow leopard is found in Pakistan, Nepal, India, Russia and in some Other countries of central Asia.

NON-NATIVE SPECIES

Species that are non-native to their environment and are **deliberately or accidentally introduced** into an ecosystem are called **non-native species.**
These are also named as invasive, alien, or exotic species.

- An introduced species is **occasionally successful** in the new environment.
- Introduction of nonnative species is a **major cause of species loss**.
- **Species are introduced** in the new Environment by **two ways intentional or Unintentional.**
-

Non-native species that have a potential to c**ause economic or environmental harm to humans** are termed as **invasive species**.

In fact, **most introduced and domesticated species** of crops and animals, such as **chickens, cattle**,and fish from around the world, are **beneficial to us**.
Grass carp was introduced in Pakistan from china.

In 1957, for example, Brazil imported wild **African bees** to help increase honey production. Instead, the bees displaced **domestic honeybees** and reduced the honey supply.

Impact of non-native species:

- Non-native species can spread rapidly if they find a new more favourable niche.
- In their new niches, these species often do not face the predators and diseases they faced before.
- They may be able to out-compete some native species in their new niches.
- Causes **habitat loss and ecosystem disturbance** .

INDICATOR SPECIES

Species that **provide early warnings of damage** to a community or an ecosystem are called **indicator species.**
For example, the presence or absence of **trout species** in water at temperatures within their range of tolerance is an indicator of water quality because trout need clean water with high levels of dissolved oxygen.

Birds are excellent biological indicators because they are found almost everywhere and are affected quickly by environmental changes such as loss or fragmentation of their habitats and introduction of chemical pesticides. The populations of many **bird species are declining**.
Butterflies are also good indicator species because their association with various plant species makes them vulnerable to habitat loss and fragmentation.

During early 1900s, **coal miners** took caged canaries (songbirds) into mines **to detect early warning signals**. These birds sing loudly and often. If they quit singing for a long period and appeared to be **distressed**, miners took this as an indicator of the presence of **poisonous or explosive gases** and got out of the mine.

Amphibians (frogs, toads, and salamanders) live part of their lives in water and part on land. Populations of various frog species can decline because of the effects of harmful factors at different points in their life cycle.

Such factors include habitat loss, drought, pollution, increased ultraviolet radiation, parasitism, disease, overhunting by humans, and non native predators and competitors.

KEYSTONE SPECIES

Some species **play a key and crucial role in maintaining the structure of a biological community**. They are called Keystone species because they **have a large effect on** the types and abundances of **other species** in an ecosystem.

The effects that keystone species have in their ecosystems is often much larger than their numbers and because of their relatively limited numbers, some keystone species are more vulnerable to extinction than others.

Keystone species can **play several critical roles in helping to sustain an ecosystem**. One such role is **pollination** of flowering plant species by **bees, butterflies, hummingbirds, bats, and other species**.

The **monarch butterfly**, which feeds on pollen in a flower and other insects pollinate flowering plants that serve as food for many plant eaters.

Examples are the alligator, wolf, leopard, lion, and some shark species.
Ecologist Robert Paine studied the keystone role of the top-predator sea star *Pisaster ochraceus* in an intertidal zone community.

Sharks are also a keystone species because they remove injured and sick animals from the ocean, and thus play an important ecological role.

The loss of a keystone species can lead to **population crashes** and **extinctions** of **other species** in a community that depends on it for certain services. This explains why it so important for scientists to identify and protect keystone species.

FOUNDATION SPECIES

Foundation species plays a **major role in shaping communities**. They create and Enhance the habitats in ways that provides **benefits to other species**. They facilitate the survival and reproduction of other species in the community.

For example, **elephants** push over, break, or uproot trees, creating forest openings in the grasslands and woodlands of Africa.
Beavers are another good example of a foundation species. Acting as "ecological engineers," they build dams in streams to create ponds and other wetlands used by other species.
Some **bat and bird** foundation species help to regenerate deforested areas. They spread fruit plants by depositing plant seeds in their droppings.

Foundation vs keystone species
A foundation species **creates or provide a foundation** of an ecosystem while keystone species **regulates and maintains that ecosystem**.

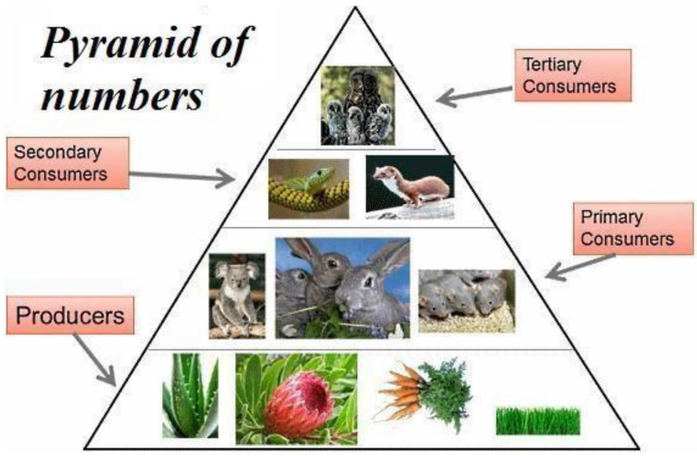
ECOLOGICAL PYRAMIDS

An ecological pyramid is **a graphical relationship among different trophic structures or levels** in an ecosystem.

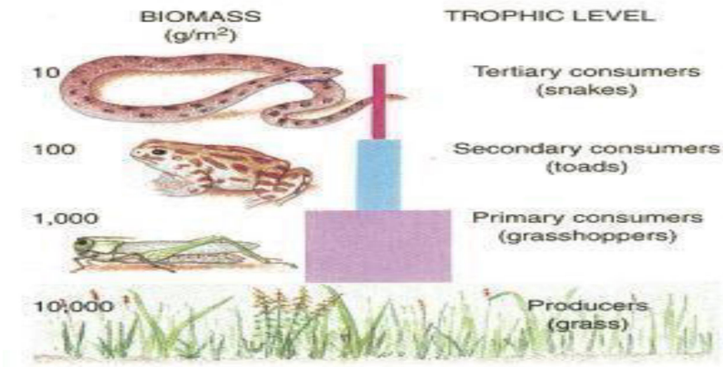
Trophic structure means the **different feeding levels** of organisms.
An ecological pyramid is further classified into three main types,

1. **Pyramid of number** 2. **Pyramid of biomass** 3. **Pyramid of energy**

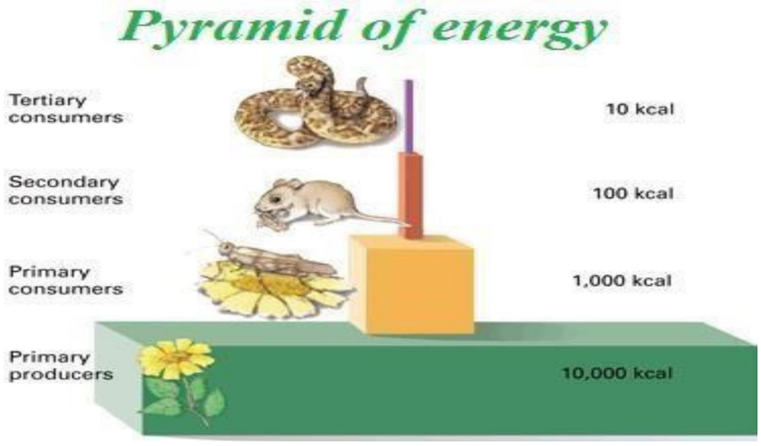
1. **Pyramid of number**
It represents the **number of individuals at each trophic level** of an ecosystem. **More number** of individuals **at base** whiles a gradual decrease in number towards top of pyramid.



2. Pyramid of biomass
It indicates mass of living matter at each trophic level of food chain. **High biomass** at the **base** while lower at top of the pyramid.



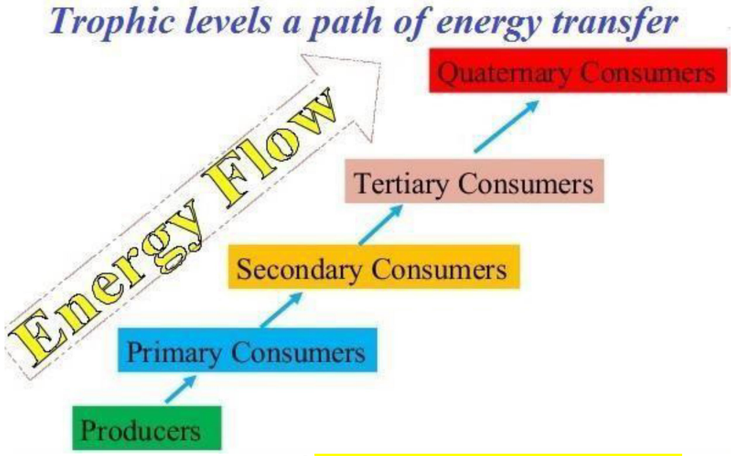
3. Pyramid of energy
Determines the **flow of energy through each trophic level** in a foodchain.



FOOD PYRAMIDS

Energy supports the activities of all organisms in the ecosystem.
A food pyramid **is a graphical representation of flow of energy through different trophic levels of an ecosystem**.

The **highest level an organism occupies** on a food chain is termed as **trophic level.**



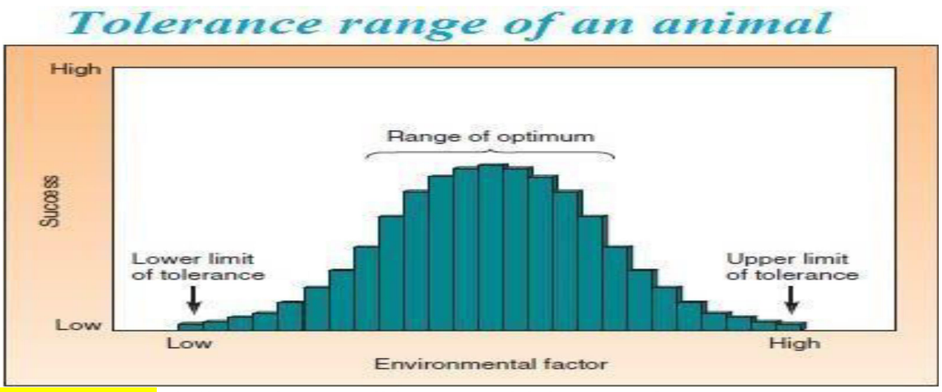
A **food pyramid** is basically consists of **producers and consumers** that belongs to different life forms. Energy in the form of food is transferred from one organism to other through different trophic levels.

ORGANISMS AND ENVIRONMENT

Actual place of an organism where it lives is called its **habitat or environment**.

An animal's habitat (environment) includes all living (biotic) and non-living (abiotic) characteristics of the area in which the animal lives.

Animals always live within a certain range of values, called the **tolerance range**, for any environmental factor.
At either limit of the tolerance range, one or more essential functions **cease**.
A certain range of values **within the tolerance range** is called the **range of optimum**.



Range of optimum defines the conditions at which **organisms are most successful** in survival.

Limiting factor

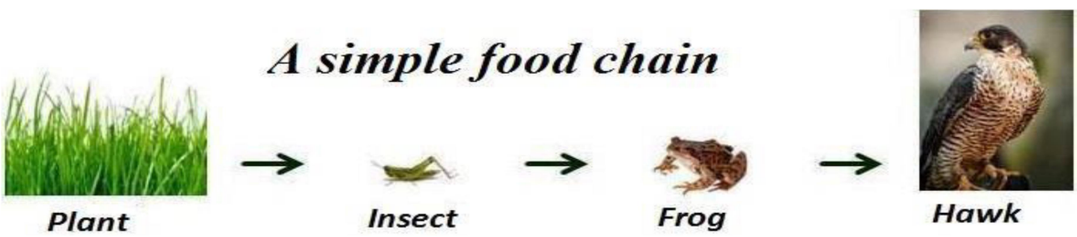
When an **abiotic factor becomes out of an organism tolerance range** then it is said to be a limiting factor. For example **o Oxygen o Water o Food**

Taxis

Response of an organism towards any abiotic factor is known as taxis. For example, a **response of organism to light** is called **photo taxis.**

FOOD CHAIN

The **sequence of organisms through which energy moves** in an ecosystem is a **food chain.** An example of a simple food chain,



Decomposers and detritus feeders consume the leaf, caterpillar, frog, and hawk after they die and return their nutrients to the soil for reuse by producers.

The chemical energy stored as nutrients in the bodies and wastes of organisms flows through ecosystems from one trophic (feeding) level to another.

A food chain determines **how chemical energy and nutrients move** from one organism to another through the trophic levels in an ecosystem.

FOOD WEB

In natural ecosystems, most consumers feed on more than one type of organism, and most organisms are eaten or decomposed by more than one type of consumer. Because of this, organisms in most ecosystems form a **complex network of interconnected food chains** called a **food web**.

Food chains and webs show **how** producers, consumers, and decomposers are **connected to one another** as energy flows through trophic levels in an ecosystem. Provide stability to ecosystem.

TROPHIC LEVELS

Because food webs are complex, so it is easy to group organisms according to the form of energy used. These groupings are called **trophic levels.**
Energy flows through these trophic levels in a food chain.

1. **Producers (autotrophs)** obtain nutrition (complex organic compounds) from inorganic materials as an energy source.

They form the **first trophic level** of an ecosystem. The most familiar producers are green plants.

2. Other trophic levels are made up of **consumers** (heterotrophs).

• Consumers eat other organisms to obtain energy.

• **Herbivores** (primary consumers) eat producers .

• Some **carnivores** (secondary consumers) eat herbivores, and other carnivores (tertiary consumers) eat the carnivores that ate the herbivores.

• **Consumers also include scavengers** that feed on large chunks of dead and decaying organic matter.

3. **Decomposers** break down dead organisms and feces by digesting organic matter extracellularly and absorbing the products of digestion.

PRODUCERS

Producers or autotrophs (**self feeders**),make the nutrients they need from compounds and energy obtained from their environment.

They can generate their own food and present the first trophic level of a food chain.
Autotrophs are divided in to **two types**
1. Photoautotrophs
2. Chemoautotrophs

1. Photoautotrophs

Most producers **capture sun light to produce energy rich Carbohydrates** (such as glucose) by **photosynthesis,** which is the way energy enters most ecosystems.

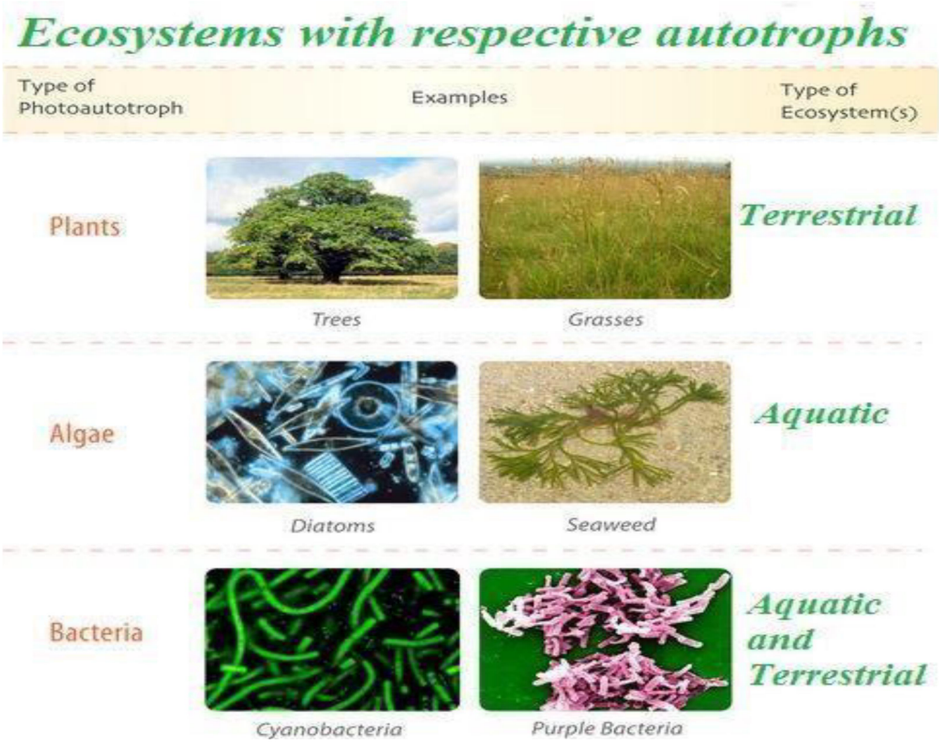
2. Chemoautotrophs

Bacteria, **can convert simple inorganic compounds from their environment into more complex nutrient compounds by using hydrogen sulphide** instead of sunlight through a process called **chemosynthesis.**
Nitrifying bacteria (Nitosomonas and Nitrobacter) are chemosynthetic in nature.

TYPES OF PHOTOAUTOTROPHS

Terrestrial producers

On land, most producers are **green plants**, which generally **capture about 1% of the solar energy** that falls on their leaves and convert it to chemical energy stored in organic molecules such as carbohydrates.
Terrestrial plants are **self supportive plants**, so they are strong and stiff in nature.
They experience **Gravity** as a main environmental force while **Wind** as an external mechanical force.
Plants are the major habitat-structuring feature in terrestrial ecosystems.



AQUATIC PRODUCERS

Organisms that **manufacture food in aquatic environment**.

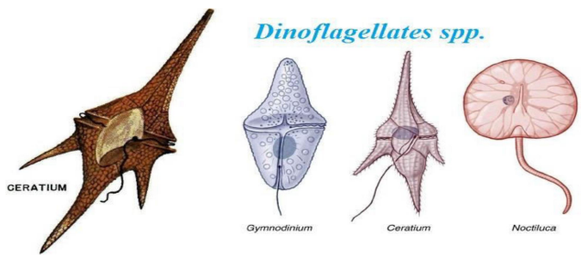
In freshwater and marine ecosystems, algae and aquatic plants are the major producers.
In open water, the dominant producers are *phytoplankton*.

Phytoplankton
Microscopic organisms that float or drift in the water
Phytoplankton, **a great source of food** for a wide range of **aquatic creatures** including whales, shrimp, snails, and jellyfish.
Phytoplankton are of **two types**

- ✓ **Diatoms**
- ✓ **Dinoflagellates**



Diatoms
o Unicellular algae
o Shells having interlocking parts.
o Major producers of an aquatic system



Dinoflagellates
o Unicellular
o Double flagella
o Second important producers of aquatic systems

Eutrophication
Too much nutrients causes harmful algal blooms (**HABs**) which are extremely toxic to living creatures (fish, shellfish, birds and mammals).



Red tide

The harmful and toxic blooms of red causes the **colour of sea water** to become **red**.

